

TANMAY WARKE

AI/ML Engineer | Data Analytics & Python Developer | Full-Stack Developer

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PROFILE

Third-year B.Tech Computer Science Engineering student (AI & Analytics) with hands-on experience across the data-to-model pipeline: cleaning, profiling, and validating datasets in Python (Pandas, Scikit-learn), then training, testing, and documenting ML systems end-to-end. Background includes validating a ~43,200-record simulated dataset, fine-tuning a model on 30,000 images, and catching critical integration bugs during hardware testing. Fluent in English and also speaks Hindi, with interest in bringing this data-quality mindset to multilingual AI datasets.

EXPERIENCE

Project-Based Intern, IEEE TechForGood

June 2026 (Remote)

IEEE Student Branch (MIT ADT University), IEEE Region 10 AIPSCC

Team: Neurox | Project: Dynamic Energy Load Balancing for Smart Buildings (RL-Based HVAC Optimization)

- Profiled and validated a simulated smart-building dataset of ~43,200 records for occupancy, environmental, and energy-consumption trends, ensuring data quality and consistency ahead of reinforcement learning training and evaluation.
- Programmed a Raspberry Pi Pico W (MicroPython) with temperature, pressure, and occupancy sensors; caught and fixed critical integration bugs, including an incorrect burden resistor value and a missing RMS conversion, before they reached the data pipeline.
- Contributed to technical proposal documentation, hardware/algorithm selection, and failure-mode analysis; ran end-to-end tests across frontend, backend, RL model, database, and hardware, achieving a 96% pass rate.
- Implemented TD3 reinforcement learning workflows using Stable-Baselines3 and PyTorch, training the agent for 100,000 episodes to optimize HVAC energy usage from simulated occupancy, temperature, and humidity data.
- Built a React.js and TypeScript dashboard integrated with backend REST APIs and PostgreSQL to visualize HVAC decisions, energy trends, and reinforcement learning outputs.

PROJECTS

Rainfall Prediction Model | Python, Pandas, Scikit-learn, Matplotlib, Seaborn

- Cleaned, explored, and engineered features from raw weather data using Pandas, then built Scikit-learn regression and classification models to predict rainfall, with results visualized in Matplotlib and Seaborn.

AI Character Detector | PyTorch, ResNet50, FastAPI, Grad-CAM, Docker

- Fine-tuned ResNet50 on a labeled dataset of 30,000 images to classify real versus AI-generated content, reaching 99.94% validation accuracy; used Grad-CAM to validate model predictions and built a FastAPI backend with Docker containerization.

EDUCATION

B.Tech in Computer Science Engineering (AI & Analytics)

2024 - 2028

MIT ADT University, Pune, India

CERTIFICATIONS

- Machine Learning with Python - IBM ([View Certificate](#))
- Introduction to AI - IBM ([View Certificate](#))
- Computer Networking - Google ([View Certificate](#))

LANGUAGES

English (Fluent), Hindi

TECHNICAL SKILLS

Data Processing & Analytics: Python, Pandas, SQL, Scikit-learn, Matplotlib, Seaborn

AI / ML: Deep Learning (PyTorch), Reinforcement Learning (TD3), Stable-Baselines3, Gymnasium

Web & Full-Stack: React.js, TypeScript, JavaScript, Tailwind CSS, Chart.js, Flask, FastAPI, Django, REST APIs

Databases & Tools: PostgreSQL, Git, GitHub, Docker

IoT & Embedded: Raspberry Pi Pico W (MicroPython), IoT Sensor Integration